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Post 12 m.y. Rotation of Southwest Washington

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Paleomagnetic field directions from the basalt of Pack Sack Lookout are compared to those from the Pomona Member of the Saddle Mountains Basalt of the Columbia River Basalt Group. The Pomona crops out over a wide region on the Columbia Plateau east of the Cascade Range, and the basalt of Pack Sack Lookout crops out well to the west of the Cascades about 30 to 60 km east of the Washington coast. Our paleomagnetic results support existing geologic and geochemical evidence that both these units are part of a single 12-m.y.-old flow that erupted in western Idaho and flowed to the Washington coast. The magnetic data further indicate that southwest Washington has undergone clockwise rotation with respect to the Columbia Plateau and stable North America. The data suggest that both a large-scale regional rotation of ~16° and locally complex small-scale rotations exist, the two being present to different degrees in the eastern and western regions of southwest Washington. The Cascade Range appears to coincide with the tectonic boundary, separating rotated and unrotated regions of Washington state. Rotation of southwest Washington appears to have been associated with the rotation of large pieces of western Oregon and southern Washington, possibly as a result of either a ridge-push force from the Basin and Range province or shear along the Pacific-North America plate boundary.

INTRODUCTION

Early and middle Tertiary rocks of western Oregon, Washington, and northern California (Figure 1) are known from paleomagnetic studies to have undergone as much as 77° of clockwise rotation relative to interior portions of North America [*Beck*, 1962; *Simpson and Cox*, 1977; *Beck and Burr*, 1979; *Beck and Plumley*, 1980; *Magill and Cox*, 1980, 1981; *Wells and Coe*, 1980; *Bates et al.*, 1981; *Magill et al.*, 1981]. Taken as a group, the paleomagnetic data from the Oregon Coast Range and western Cascade Range share three characteristics: First, the magnetic inclinations agree closely with those from rocks of similar age from stable North America. Although the error limits are large (±400 km), this agreement indicates that no measureable north-south translation has taken place. Second, the older Eocene rocks (~77° rotation) are rotated more than the younger Oligocene and Miocene rocks (~30°), as would be expected if clockwise rotation had occurred progressively or episodically from the Eocene to the present. Third, the magnitude and the clockwise sense of the rotations observed in rocks of a given age are regionally consistent. On the basis of this coherence and the apparent absence of major crosscutting tectonic boundaries, *Simpson and Cox* [1977], *Magill and Cox* [1980], and *Magill et al.* [1981] have proposed that the Oregon Coast Range and western Cascade Range have rotated as an essentially rigid plate.

On the basis of the available data, *Bates et al.* [1981] suggest that rotation occurred at a nearly constant rate from Eocene time to the present. Alternatively, *Magill et al.* [1981] and *Magill and Cox* [1980] propose two distinct periods of rotation, phase I during the Eocene accretion of the Oregon Coast Range to the continent, and phase II during post early Miocene extension of the Basin and Range province.

The existing paleomagnetic data leave two important questions unanswered about the history of late Cenozoic rotation. First, did rotation of the western Cascades and adjoining terrane continue through the Neogene, as would be expected if it is associated with extension of the Basin and Range province, or was it completed at some earlier time? Second, where is the tectonic boundary separating the rotated western Cascades from the unrotated regions to the east? Eastern Washington and northeastern Oregon can probably be considered to be part of unrotated stable North America throughout the Neogene because Miocene basalts of the Columbia River Basalt Group of extreme southeastern Washington and northeastern Oregon [*Watkins and Baksi*, 1974; *Choiniere and Swanson*, 1979; *Hooper et al.*, 1979] are rotated only slightly if at all (indicated rotations are 3° ± 12°, −1° ± 16°, 1° ± 13° if statistics outlined by *Beck* [1980] are applied to the respective data). However, exactly where the tectonic boundary lies between the rotated western Cascades and the unrotated terrane to the east is unknown. Also unknown is the nature of the boundary. Is it diffuse or sharp?

To answer these questions, we studied the paleomagnetism of late middle Miocene basalts of southern Washington, which extend from the Idaho border to the Pacific coast (Figures 2 and 3). The great areal extent of these Miocene basalts provides an opportunity to improve the temporal resolution of post middle Miocene rotation and also shed some light on the size of the rotated domains and the location of important tectonic boundaries.

DESIGN OF EXPERIMENT

Our experimental approach was to analyze samples from a single basalt flow over a broad region (500 × 150 km) [*Swanson et al.*, 1979, Plate 1] from the Idaho-Washington border to the Pacific coast. We assume that when this flow cooled, it faithfully and 'instantaneously' recorded a well-behaved geomagnetic field across the entire region covered

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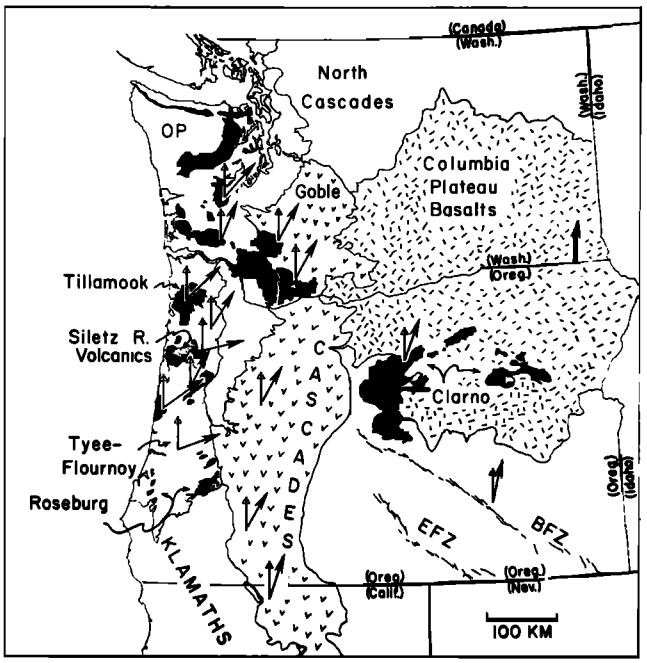


Fig. 1. Generalized geotectonic map of Oregon and Washington, Arrows indicate the mean paleomagnetic directions found for the respective geologic units. The open arrow marks the expected declination and the solid arrow the observed declination. The arrows have been adjusted so that the expected declinations point due north. An easterly directed solid arrow therefore indicates at 90° clockwise rotation. Goble, Goble Volcanics; Tillamook, Tillamook Volcanics; Tyee, Tyee Formation; Flournoy, Flournoy Formation of *Baldwin* [1974]; Roseburg, Roseburg Formation of *Baldwin* [1974]; Clarno, Clarno Formation; OP, Olympic Peninsula; BFZ, Brothers fault zone; EFZ, Eugene-Denio fault zone. From *Magill et al.* [1981, Figure 2].

by the flow. This assumption should be valid if the cooling time for the flow was no more than a few decades and the field was essentially dipolar. Small intraflow variations in direction are to be expected from spatial variations in the earth's dipole and nondipole fields and from local site-specific magnetic anomalies present at the time of cooling. In parts of the flow that have undergone post-cooling tectonic rotation, paleomagnetic directions will be rotated relative to less rotated or unrotated parts of the flow. The paleomagnetic directions thus provide a direct measure of differential rotation over the region sampled. In conventional studies of multiple flows, geomagnetic secular variation requires that 30 to 40 distinct flows be sampled in order to determine the tectonic rotation with an accuracy of 10° to 15°. With the single-flow technique, the differential rotation can be determined from one flow with an accuracy limited only by intraflow variations in the direction of magnetization within undeformed parts of the flow.

GEOLOGY

The exposed source vents for the Columbia River Basalt Group are located in the eastern half of the Columbia Plateau near the Idaho border [*Swanson et al.*, 1979] (Figure 3). These tholeiitic basalts erupted as sheet flows that flowed westward through a broad downwarp in the region of the present Columbia River [*Lowry and Baldwin*, 1952; *Waters*, 1961; *Snavely and Wagner*, 1963; *Peck et al.*, 1964]. Some flows originating in the eastern Columbia Plateau flowed into the Willamette Valley between the Oregon Coast Range and Cascade Range [*Peck et al.*, 1964; *Snavely and Wagner*, 1963; *Beeson et al.*, 1979], while a few flows proceeded westward, entered the Pacific Ocean, and produced pillow

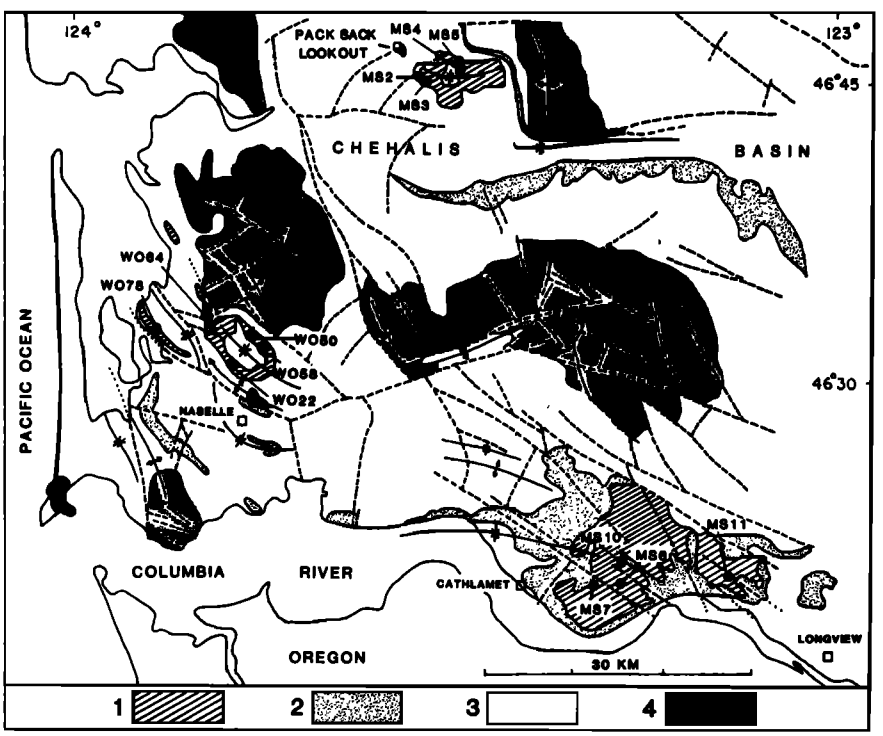


Fig. 2. Simplified geologic map of southwest Washington with sample localities. 1, upper middle Miocene basalt of Pack Sack Lookout; 2, middle Miocene Wanapum and uppermost part of the Grande Ronde Basalts; 3, Miocene, Oligocene, and Eocene sedimentary rocks; 4, Eocene Crescent Formation. Dashed lines are faults. Geology from *Pease and Hoover* [1957], *Wagner* [1967], and *Wells* [1979, 1981].

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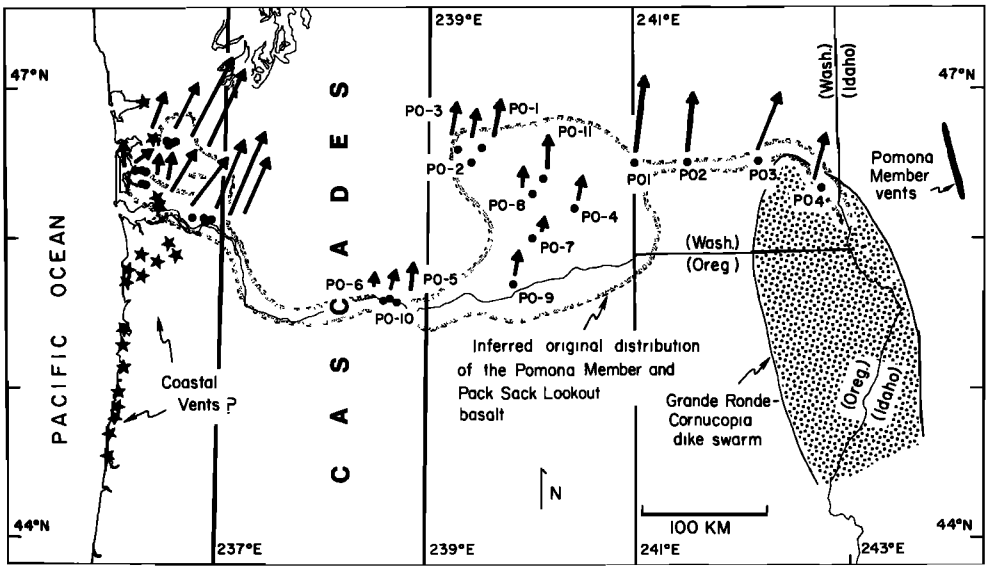


Fig. 3. Simplified geologic map of northern Oregon and southern Washington. Paleomagnetic sampling sites in the Pomona Member and the basalt of Pack Sack Lookout are shown by arrows, the azimuth determined by the measured paleomagnetic declination (lower hemisphere projection) and the length inversely proportional to the radius of the 95% confidence circle. See Figure 2 for identification numbers of the Pack Sack Lookout sample localities. The inferred original distribution of the Pomona and Pack Sack Lookout basalts is modified from *Swanson et al.* [1979]. The Grande Ronde–Cornucopia dike swarm is the general source region for much of the Columbia River Basalt Group, location from *Snavely et al.* [1973]. Location of the Pomona vents are from *Swanson et al.* [1979] and *Camp* [1981].

lava and breccia in northern Oregon and southern Washington [*Snavely et al.*, 1973] nearly 500 km west of their source.

Although a few flows of the Columbia River Basalt Group are generally recognized as reaching the coast, it is uncertain whether or not at all the Miocene tholeiitic basalts cropping out in coastal northern Oregon and southern Washington (Figure 3) are plateau-derived Columbia River basalts. *Snavely et al.* [1973] have shown that three coastal basalts, the Depoe Bay Basalt, the Cape Foulweather Basalt and the basalt of Pack Sack Lookout, are coeval with and geochemically identical to flows in three units of the Columbia River Basalt Group, the Grande Ronde Basalt, the Frenchman Springs Member of the Wanapum Basalt, and the Pomona Member of the Saddle Mountains Basalt, respectively. *Choiniere and Swanson* [1979] cite paleomagnetic evidence to support these correlations. However, the local intrusive character of the coastal units led *Snavely et al.* [1973] to propose that local volcanic centers were the sources for these coastal basalts. Alternatively, *Beeson et al.* [1979] suggest that the coastal basalts are the westernmost exposures of the plateau-derived basalts and that dikes and sills along the coast are not associated with local vents but are the result of invasion of dense basalt flows into soft, buoyant estuarine and deltaic sediments.

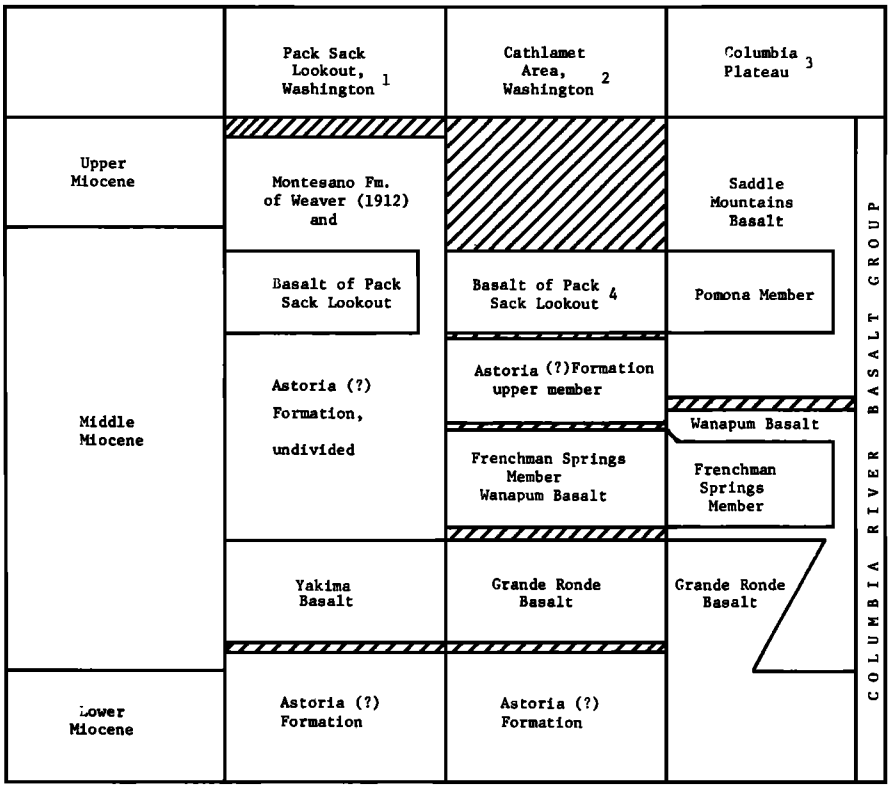
The flow that we examined is the Pomona Member of the Saddle Mountains Basalt of the Columbia River Basalt Group on the Columbia Plateau [*Schmincke*, 1967] and its apparent western extension, the basalt of Pack Sack Lookout [*Snavely et al.*, 1973] of southwestern Washington (Figures 2 and 3). Our new paleomagnetic results are from outcrops of the Pack Sack Lookout basalt west of the Cascades. The Pomona flow has been previously sampled in the western [*Rietman*, 1966] and eastern Columbia Plateau [*Choiniere and Swanson*, 1979]. Present exposures of the Pomona and the Pack Sack Lookout basalts are limited to the Columbia Plateau and Cascade Range (Pomona) and Washington Coast Range (Pack Sack Lookout). No outcrops are known to occur in the intervening gap.

The Pomona flow appears to have issued from vents in western Idaho [*Swanson et al.*, 1979; *Camp*, 1981] and to have flowed westward, in places as an intracanyon flow and in other places as a broad sheet flow. We note that in several localities on the plateau, two Pomona flows are reported. However, at these localities the basalt commonly invades sediments and may represent separate local lobes of a single flow [*Swanson et al.*, 1979]. At present the weight of evidence seems to indicate that only a single Pomona flow existed. This conclusion is being tested by paleomagnetic sampling at sites where two flows are reported (R. E. Wells et al., work in progress, 1982).

In the Washington Coast Range, the basalt of Pack Sack Lookout occurs as a flow at the top of a sequence of undisputed Columbia River basalt flows along the north side of the Columbia River near Cathlamet and as a flow above Miocene marine sediments near Pack Sack Lookout 60 km to the north (Figure 2). At these localities the Pack Sack Lookout flow unconformably overlies shallow marine and fluvial sediments. The present-day outcrops appear to be remnants of a flow that bifurcated around a topographic high near Longview, then flowed as two restricted lobes along the axes of the lower Columbia River Basin and the Chehalis Basin (Figure 2). To the west, about 30 km from the present coast, no subaerial flow of the basalt of Pack Sack Lookout is found. However, Miocene basalts of Pack Sack Lookout chemistry are found in this region forming either large dikes and sills or an invasive flow that intruded, often discordantly, sediments as old as late Eocene.

The question whether the Pomona Member and the basalt of Pack Sack Lookout are actually parts of the same flow is of particular importance because we compare paleomagnetic directions from sampling localities in the Pomona and Pack Sack Lookout units on the assumption that these units have

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1. Snavely et al.,1973; this report; 2. Wells, 1981, Snavely et al., 1973; 3. Swanson et al., 1979; 4. Wells (1981) mapped these rocks as Pomona Member.

Fig. 4. Stratigraphic relations of Miocene basalts in southwest Washington and the Columbia Plateau.

preserved identical remanent magnetizations. However, if the Pomona and Pack Sack Lookout cooling times differ by a decade or more, secular variation of the geomagnetic field could result in flow magnetizations that are discordant by an amount equal to the rotations we wish to detect. We will therefore review in some detail the geologic data relevant to the correlation of the Pomona and the Pack Sack Lookout units.

Six geologic observations suggest that the Pomona Member and the basalt of Pack Sack Lookout are parts of the same flow.

1. The two basalts occupy similar positions in the regional stratigraphy (Figure 4). Except for the sills near the coast, both units are underlain by the plateau-derived Grande Ronde and Wanapum basalts [*Snavely et al.*, 1973; *Schmincke*, 1967; *Wells*, 1981], which can be firmly correlated from the coast to the Columbia Plateau. Similarly, the Pomona and the basalt of Pack Sack Lookout are both interbedded with late Miocene sedimentary rocks [*Schmincke*, 1967; *Snavely et al.*, 1973]. In the Cathlamet region, *Wells* [1981] has mapped in detail a sequence of at least seven flows of Grande Ronde Basalt and Wanapum Basalt of the Columbia River Basalt Group that covers about 600 km^2. The Pack Sack Lookout basalt is the uppermost flow of the sequence. On the basis of field and geochemical criteria, these basalt flows exhibit a stratigraphic order corresponding to that observed in the more numerous flows of the Columbia Plateau. This correspondence is strong circumstantial evidence for a plateau source for the Pack Sack Lookout basalt in the Cathlamet region.

2. *Anderson* [1980] has mapped the Pomona Member crossing the axis of the Cascades (Figure 3) as an intracanyon flow in the canyon of the ancestral Columbia River. Anderson proposed that this canyon was a conduit that enabled the Pomona to cross the Cascades and flow over the coastal plain of southwest Washington. He interprets the basalt of Pack Sack Lookout in the Cathlamet region (Figure 2) as an extension of the Pomona flow. We suggest that the basalt of Pack Sack Lookout is a remnant of a separate flow lobe that was channeled into the Chehalis Basin.

3. K-Ar age determinations yield a late middle Miocene date for the Pomona (12 m.y.; *Mckee et al.* [1977]) and a late Miocene date for the basalt of Pack Sack Lookout (9.0 ± 1.4 m.y. on a single sample; see *Snavely et al.* [1973]). These ages are permissive though not strongly supportive of the units being parts of the same flow.

4. The major element and trace element geochemistry [*Snavely et al.*, 1973; *Bowman et al.*, 1974] indicate that the Pomona and Pack Sack Lookout basalts are chemically identical. Figure 5 [from *Bowman et al.*, 1974] shows chemical analyses for 25 elements obtained from the Pack Sack Lookout basalt, the Pomona basalt, and five other plateau-derived units. The Pack Sack Lookout analyses agree with only one plateau unit, the Pomona, and the agreement is excellent. This is an important correlation considering the wide chemical variability within the Columbia River Basalt Group. *Snavely et al.* [1973] and *Bowman et al.* [1974] conclude that the Pomona and pack Pack Lookout basalts are probably the product of a common magma source.

5. The lead and strontium isotope ratios are indistinguishable for the two basalt units [*Snavely et al.*, 1973].

6. The Pomona and Pack Sack Lookout basalts share the

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same distinctive mineralogy and petrographic textures. Both units contain abundant small phenocrysts and glomerophenocrysts of labradorite, augite, and olivine with many labradorite phenocrysts containing glass and pyroxene inclusions [*Snavely et al.*, 1973]. These petrographic criteria are diagnostic and serve in the field as a basis for distinguishing the Pomona and Pack Sack Lookout from other plateau and coastal flows. The similarity between the Pomona Member and Pack Sack Lookout basalt is impressive on all scales, from regional stratigraphy down to the composition of individual phenocrysts. We believe that these characteristics establish the basalt of Pack Sack Lookout as a plateau-derived western extremity of the Pomona Member.

PALEOMAGNETIC DATA

Our sampling localities within the Pack Sack Lookout basalt are at four sites along the Washington side of the Columbia River between Cathlamet and Longview, Washington, four sites near Pack Sack Lookout, and five sites about 7 km north of Naselle, Washington. From 6 to 13

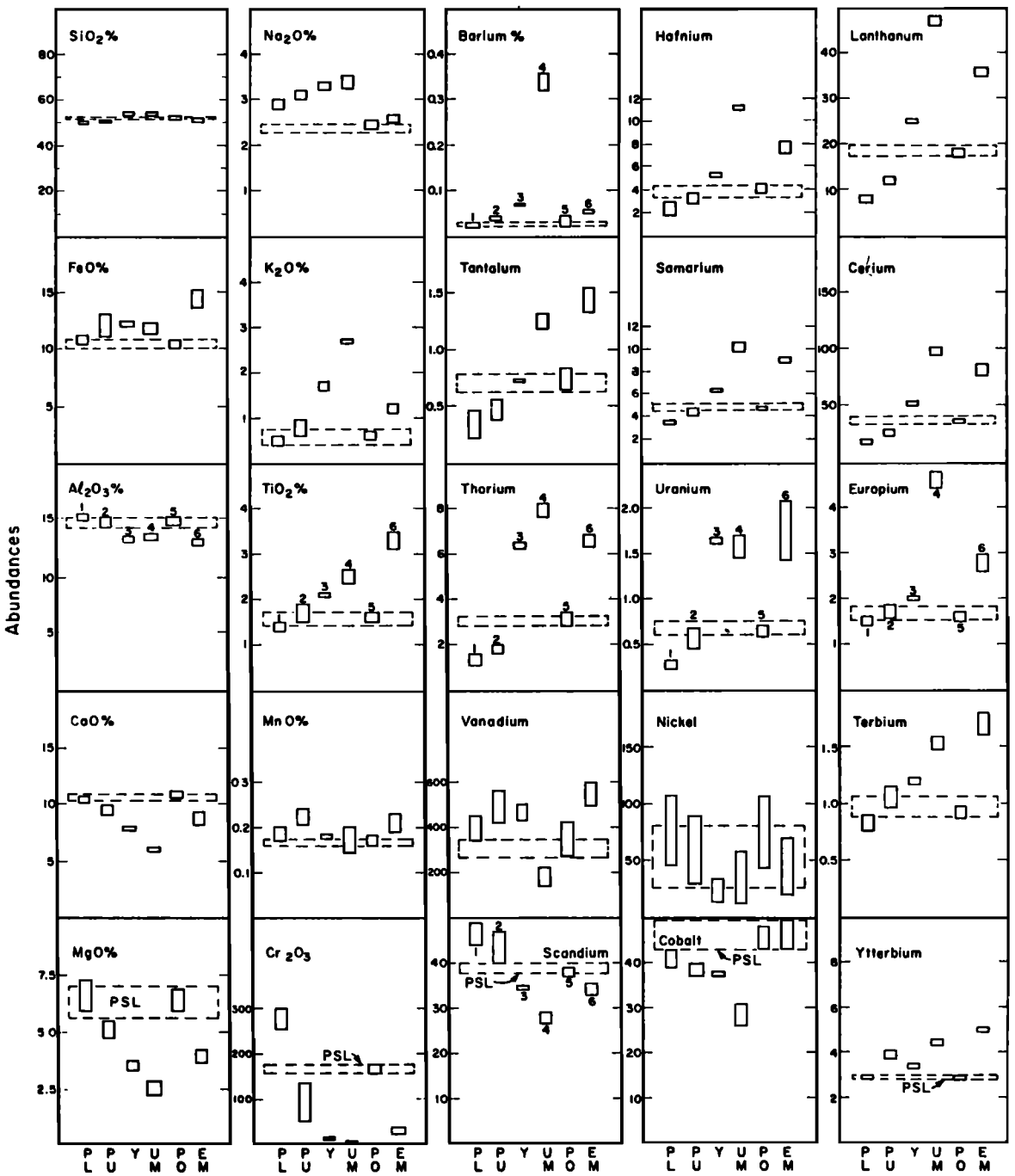


Fig. 5. Chemical compositions of flows of the Columbia River Basalt Group (small rectangles): 1, Picture Gorge, lower part; 2, Picture Gorge, upper part; 3, Grande Ronde Basalt of Yakima Basalt subgroup; 4, Umatilla; 5, Pomona Member of Saddle Mountains Basalt; 6, Elephant Mountain Member of Saddle Mountains Basalt; 7, basalt of Pack Sack Lookout (PSL) is shown as a long horizontal dashed rectangle. Only the Pomona Member matches the basalt of Pack Sack Lookout for all elements. Abundances are in ppm where not indicated. Figure from *Bowman et al.* [1974].

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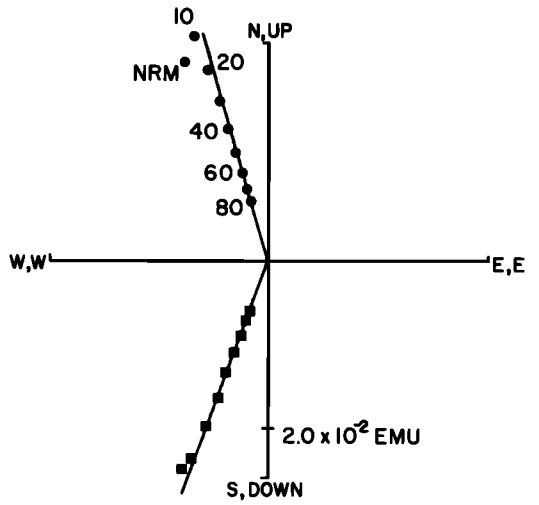


Fig. 6. Zijderveld component plot of sample MS10-1-1 during stepwise alternating field demagnetization. The squares represent the west and south components and the dots the west and up components of magnetization. The peak demagnetization values are shown in milliteslas. The lines show the expected demagnetization path for the removal of a single component of magnetization. Directions have not been corrected for tectonic tilt.

samples spaced 1 to 5 m apart along an outcrop were collected from each of the 13 sites. The individual samples were oriented with a sun compass where possible or with a magnetic compass corrected for local magnetic anomalies by means of backsighting techniques. Each sample was magnetically cleaned in a stepwise progression with alternating peak fields ranging from 5 to 80 mT. Zijderveld component plots were used to assess the directional stability and coercivity spectrum of each sample. The paleomagnetic results were of high quality, and Figure 6 is typical of our results. The straight demagnetization trend intersecting the origin is diagnostic of the removal of a single component of magnetization. All samples were subjected to demagnetization until such a straight demagnetization trend developed, the trend being defined by a minimum of four demagnetization steps.

None of the samples was found to be directionally unstable. However, the direction of one sample from each of three sites (MS6, WO58, and WO64) was rejected because it lay greater than two angular standard deviations (2 σ) from the overall directional grouping of the respective site. The average α_{95} (radius of 95% confidence) for the 13 Pack Sack Lookout sites was 2.6°, with none larger than 4.0°. The four sites collected in the region of Pack Sack Lookout span both limbs of a syncline and pass a fold test upon unfolding. Tilt corrections for these sites were derived from attitudes of underlying sedimentary rocks mapped by *Pease and Hoover* [1957] and *Wagner* [1967]. These corrections were deemed more reliable than local indicators of tilt observed at the sampling sites. We note that although tilt corrections are the largest source of uncertainty in the measurements reported here, the average results do not change appreciably for the full range of possible tilt corrections (see the appendix). The dips are all less than 12°, yet the precision parameter increased from 154 to 553 upon unfolding. This constitutes a positive fold test with 90% confidence. The positive fold test indicates that the measured magnetization predates the formation of the syncline and probably is the primary thermoremanent magnetization. The tilt corrections for the southern sites near the Columbia River were derived from the regional topographic slope where Miocene basalt flows cap the topography and dip less than 2° to the northeast on the south flank of a gentle syncline [*Wells*, 1981].

The field directions of sites WO22, WO50, WO58, WO64, and WO78 near Naselle, Washington, are discordant by as much as 27° (average discordance 19.3°) from the average direction of the other Pack Sack Lookout sites (Tables 1 and 2). This discordance is very large considering the 4° angular standard deviation of the other Pack Sack Lookout sites and is somewhat unexpected because the Naselle sites pass a fold test with k increasing from 9 to 42 upon unfolding. However, even after tilt correction, the scatter between the Naselle sites is much larger than that of the other sites (see interpretation section). The mineralogy, geochemistry [*Snavely et al.*, 1973], and general paleomagnetic field directions appear to indicate that these sites are in Pack Sack Lookout basalt. These basalt sills may represent invasions of the Pack Sack Lookout flow into the sedimentary section, although an intrusive origin from below can not be disproved on the basis of the available geologic and paleomagnetic evidence. The scatter and directional discordance have four likely sources. First, these western sites are in intrusive bodies with a thickness locally in excess of 300 m. The cooling time for such a body is presumably much longer than that of the thinner surface flow of the Pack Sack Lookout basalt. Slow cooling introduces the possibility that the aberrant directions may record a change in the direction of the geomagnetic field during cooling. Second, intrusion of basalt into sediments, either from above or below, may have disrupted bedding so that the tilt corrections applied (all derived from the sediments) may be incorrect. Third, post 12-m.y. folding and faulting may have been complex in this region [*Wells*, 1979], resulting in unrecognized tilting about axes other than the strike of the sediments. The large tilt corrections for these sites (23° to 41°), suggest that significant unrecognized tilting may have occurred. Fourth, the basalt sills crosscut a regional unconformity and conformably intrude late Eocene rocks on the flank of a large basement uplift (Figure 2). Sites in this area may have been overcorrected for the tilt of underlying Eocene rocks that were involved in an earlier period of folding. In fact, after tilt correction, the three sites with anomalously steep inclinations relative to the other Pack Sack Lookout sites are those sites where the basalt intrudes Eocene sedimentary rocks. If these sites are corrected for tilt along the observed local strikes but with only enough dip to yield an inclination similar to the other Pack Sack Lookout sites, then, with one exception, a good grouping of directions close to the declination of the other Pack Sack Lookout sites is produced. The exception is site WO64, which lies at the junction of two faults. Its divergence in declination is matched by the difference between the local strike of bedding compared to neighboring sites and suggests local rotation of a small block a few kilometers across.

Paleomagnetic results for 11 sites from the Pomona Member were reported by *Rietman* [1966], and results for another 4 sites by *Choiniere and Swanson* [1979]. It is important to assess the accuracy of these studies before attempting to interpret the results. From 5 to 8 samples were collected from each site and all samples were magnetically cleaned using alternating field demagnetization. Selected samples were subjected to thermal demagnetization. The accuracy of the magnetic results can best be judged from the precision

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TABLE 1. Compilation of Paleomagnetic Data From the Pomona Member and Basalt of Pack Sack Lookout

Site	Location		Attitude					Field Direction		Field Direction*		
	Latitude, deg	Longitude, deg	Strike, deg	Dip, deg	N	k	α_{95}	I, deg	D, deg	I, deg	D, deg	ΔD
Pack Sack Lookout												
MS2	46.72	236.46	152	7NE	12	812	1.5	−54.0	204.9	−53.9	205.6	2.5
MS3	46.69	236.50	313	12NE	13	119	3.8	−49.0	200.3	−48.9	201.0	5.8
MS4	46.72	236.50	29	7SE	12	833	1.5	−50.4	209.1	−50.3	209.8	2.4
MS5	46.73	236.53	35	4SE	12	229	2.9	−47.4	207.4	−47.3	208.1	4.3
MS6	46.20	236.78	330	2NE	12	446	2.1	−46.1	202.9	−46.5	203.6	3.0
MS7	46.20	236.72	330	2NE	12	328	2.4	−49.0	217.8	−49.4	218.5	3.6
MS10	46.22	236.76	330	2NE	12	707	1.6	−49.1	203.9	−49.5	204.6	2.4
MS11	46.21	236.91	0	0	6	1014	2.1	−51.2	203.9	−51.6	204.5	3.4
WO22	46.43	236.20	310	23NE	10	311	2.6	−53.5	211.4	−53.8	212.2	4.4
WO50	46.47	236.22	100	28S	8	283	3.3	−72.9	233.0	−73.1	233.8	11.3
WO58	46.44	236.23	135	29SW	5	440	3.7	−65.2	191.0	−65.5	191.8	8.9
WO64	46.47	236.16	165	41W	6	671	2.6	−75.6	189.2	−75.8	190.0	10.5
WO78	46.46	236.10	320	25NE	7	229	4.0	−60.9	174.9	−61.2	175.7	8.3
Pomona												
PO-1	46.71	239.53			8	235	3.6	−52.7	192.8	−52.2	192.6	5.9
PO-2	46.62	239.43			7	161	4.8	−52.1	191.9	−51.8	191.7	7.8
PO-3	46.69	239.35			6	209	4.6	−47.8	191.9	−47.4	191.7	6.9
PO-4	46.27	240.43			6	146	5.6	−50.1	193.3	−50.0	192.8	8.8
PO-5	45.69	238.65			7	175	4.6	−54.4	185.7	−55.2	185.8	7.9
PO-6	45.69	238.59			6	26	13.3	−51.8	186.2	−52.6	186.4	21.8
PO-7	46.13	240.00			6	39	10.9	−46.7	192.9	−46.8	192.6	16.0
PO-8	46.42	240.02			8	48	8.1	−48.4	185.8	−48.2	185.4	12.3
PO-9	45.80	239.80			8	104	5.5	−48.8	191.7	−49.3	191.5	8.4
PO-10	45.71	238.63			5	104	7.5	−51.5	197.7	−52.2	197.8	12.1
PO-11	46.54	240.13			7	250	3.8	−52.4	182.1	−52.0	181.7	6.2
PO1	46.59	241.00	0	0	6	1500	1.7	−51.9	187.8	−51.4	187.1	2.8
PO2	46.59	241.45	0	0	6	900	2.2	−53.4	186.3	−52.8	185.5	3.7
PO3	46.65	242.23	0	0	6	1100	2.0	−54.6	203.4	−53.8	202.4	3.5
PO4	46.41	242.82	80	5S	6	500	2.9	−50.6	195.4	−50.0	194.2	4.6

N, number of samples; k, precision parameter; α_{95}, radius of 95% confidence; I, inclination; D, declination; ΔD, declination uncertainty at 95% confidence where $\Delta D = \sin^{-1}\{\sin(\alpha_{95})/\cos I\}$. References for sites MS2–MS11 and WO22–WO78, this paper; sites PO-1–PO-11, *Rietman* [1966]; sites PO1–PO4, *Choiniere and Swanson* [1979].

*Field direction referenced to a point central to all sites 46.4°N, 238.8°E where the regional changes of inclination and declination have been computed from the average VGP of the Pomona sites at 73.5°S, 208.2°E.

parameter k and α_{95} for the respective sites. The α_{95} statistics for the 4 sites of *Choiniere and Swanson* [1979] range from 1.7° to 2.9° and k ranges from 186 to 203. A tilt correction of 5° was applied to one site, PO4 (Table 1). The overall quality of *Choiniere and Swanson*'s [1979] results appears high; only a negligible amount of uncertainty is contributed by the single small tilt correction. The data of Rietman are of lower accuracy, largely because samples from 5 sites were collected as oriented hand samples rather than by in situ drilling. The average α_{95} was 6.6°, ranging from 3.6° to 13.3°, with k ranging from 26 to 250. The sites sampled by hand exhibit the worst directional grouping. Neither the uncorrected directions nor the tilt corrections are listed by Rietman, and their absence leads to some uncertainty about the errors introduced by tilt corrections. The tilt corrections are thought to be less than 10° (J. D. Rietman, oral communication, 1981), and the sites pass a fold test using the criterion that the precision parameter increases upon unfolding. The fold test is important because it suggests that the tilt corrections were not significantly in error.

The individual site directions have probably recorded small regional variations in the direction of the geomagnetic field at the time the flow cooled. Since such regional variations in magnetic direction are of no tectonic significance, an estimate of the maximum magnitude of these

TABLE 2. Average Paleomagnetic Directions From the Pomona Member and Basalt of Pack Sack Lookout

	N	k	α_{95}	I, deg	D, deg	ΔD	ΔI
Pack Sack Lookout	13	47	6.1	−56.4	205.0	11.1	6.1
Pack Sack Lookout (without WO22–WO78)	8	372	2.9	−49.8	207.0	4.5	2.9
Pack Sack Lookout (without WO22–WO78, MS7)	7	660	2.4	−49.7	205.3	3.7	2.4
Pomona	15	393	1.9	−51.2	190.6	3.1	1.9
Pomona (without PO3)	14	498	1.8	−50.9	189.8	2.8	1.8

N, number of sites; k, precision parameter; α_{95}, radius of 95% confidence; I, inclination; D, declination; ΔD uncertainty in declination at 95% confidence, where $\Delta D = \sin^{-1}\{\sin(\alpha_{95})/\cos I\}$; ΔI, uncertainty in inclination at 95% confidence where $\Delta I = \alpha_{95}$.

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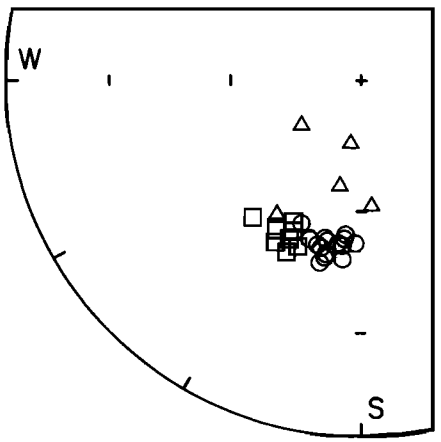


Fig. 7. Paleomagnetic field directions for sites from the Pomona Member (circles, $N = 15$) and Pack Sack Lookout basalt (squares are surface flows, triangles are sills, $N = 13$) corrected for tectonic tilt. Directions are all of reversed polarity and plot on the upper hemisphere of this stereographic plot.

variations is important. The paleofield variation consists of dipole and nondipole components. The dipole component depends upon the locations of the various sites and the orientation of the paleodipole relative to the rotation axis. Near the pole, closely spaced sites could have declinations differing by 180°. The dipole variation was removed from the data by referencing each field direction to a common central location (maximum adjustment <1°, Table 1) using the average virtual geomagnetic pole (VGP) of the Pomona sites as the best estimate of the paleodipole. This same adjustment computed using the 20 m.y. pole of *Irving* [1979] produces similar results, the two methods having a maximum difference of <1°. Although the nondipole variation cannot be determined, its magnitude was estimated by analyzing the nondipole geomagnetic field of 1965 (IGRF, *Nagata* [1969]). Between latitudes 55°N and 55°S, the largest observed declination difference for locations 300 km apart (sampled at 1° intervals around circles of latitude) was 11°. The average difference was 0.9°, and 95% of all differences were less than 3.5°. These differences provide an estimate of the uncertainty introduced by the nondipole paleofield under the assumption that the ratio of the nondipole to the dipole field was the same in 1965 as it was when the flow cooled. A nondipole paleofield vastly larger than the present seems unlikely because the Pomona VGP is displaced only 19° from the Miocene paleomagnetic pole for stable North America [*Irving*, 1979]. This displacement is fairly typical for a field with the present ratio of nondipole to dipole components [*Cox*, 1970].

Figure 7 and Tables 1 and 2 summarize the mean field directions of the basalt of Pack Sack Lookout and the Pomona Member [*Rietman*, 1966; *Choiniere and Swanson*, 1979]. The overall average field direction computed from the 13 sites of Pack Sack Lookout basalt is I (inclination) = −56°, D (declination) = 205°. Rather poor directional grouping is indicated by the precision parameter, $k = 47$ (α_{95} = 6.1°). The directional dispersion is largely due to the five invasive sites (WO22, etc.). Exclusion of these sites because of their dispersion results in a similar average direction but much better grouping, $I = -50°$, $D = 207°$, $k = 372$, α_{95} = 2.9°. The direction of one site (MS7) is still more than 2 σ away from this overall mean. Although the outcrop of MS7 has been struck by lightning, this does not explain the anomaly because no large secondary magnetic components were found except in the immediate region of the lightning strike. Small anomalous components of high coercivity (>80 mT) were found in four samples, yet excluding these samples does not remove the anomaly. A more likely explanation of the anomalous direction is that MS7 lies between a mapped fault and a suspected fault about 1 km apart, suggesting that the area between the two faults may have rotated as a small block. Excluding site MS7 from the overall average because of its directional discordance yields our best estimate of the Pack Sack Lookout field direction, $I = -50°$, $D = 205°$, α_{95} = 2.4°.

Paleomagnetic data of the Pomona basalt exhibit a direction similar to the Pack Sack Lookout direction, $I = -51°$, D = 191°. Again the field direction of one site (PO3) lies more than 2 σ away from the overall mean. *Choiniere and Swanson* [1979] report nothing anomalous about the magnetization of PO3, yet the site is from an intracanyon segment of the Pomona flow within the ancestral Snake River canyon (D. A. Swanson, oral communication, 1981). This introduces the possibility that the Pomona flow cooled in a local anomalous magnetic field associated with the walls of the ancestral canyon. Additionally, secular variation may also have affected the site due to slow cooling of the thick intracanyon flow [*Choiniere and Swanson*, 1979]. Excluding PO3 yields our best estimate of the Pomona field direction, I = −51°, D = 190°, α_{95} = 1.8°.

The declination data are shown in Figures 8 and 9 plotted along a line normal to the regional structural grain to detect possible geographic domains or trends. Trends are most easily seen in the weighted averages of neighboring sites

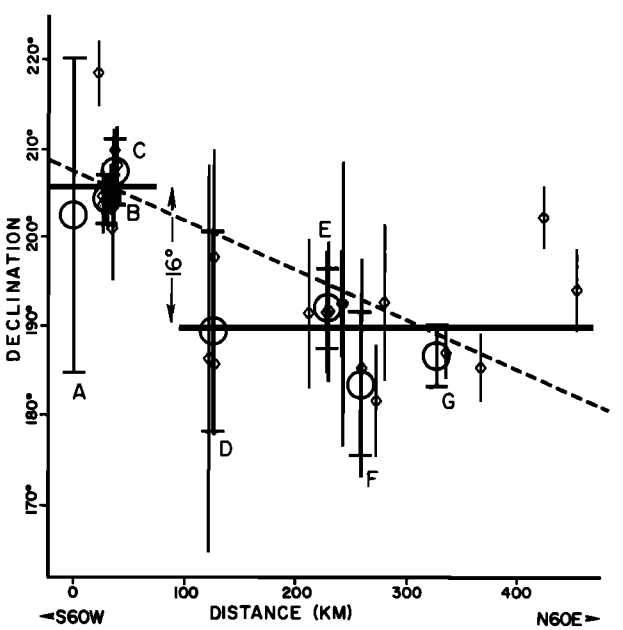


Fig. 8. Paleomagnetic declinations (diamonds) of the Pomona–Pack Sack Lookout flow with site locations projected onto a line trending N60°E. Error bars for the diamonds are ΔD values listed in Table 1. Regional averages are shown (circles, labeled A through G) as listed in Table 3, error bars are the σ_μ or S_μ of Table 3, whichever is larger. Solid line shows the favored rigid plate model with two domains, a rotated Pack Sack domain, and an unrotated plateau domain. The dotted line depicts the regional shear model with a line fit to the data showing a progressive increase in rotation toward the west.

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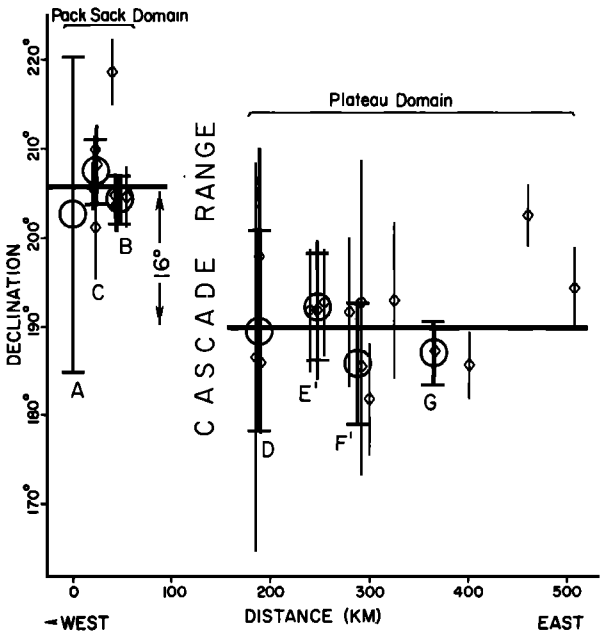


Fig. 9. Paleomagnetic declinations of the Pomona–Pack Sack Lookout flow with site locations projected onto an east-west line. Symbols and error bars as in Figure 8.

(circle symbols, Figures 8 and 9, Table 3) for which site declinations were weighted by their respective standard deviations. The 90% confidence limits were computed as noted in Table 3. Weighted averages are particularly useful in analyzing Rietman's Pomona data, which are characterized by neighboring sites of variable precision. Averaging enhances regional trends in the data by reducing scatter caused by undetected or improperly corrected local folds and faults or other random errors. The five sites near Naselle are represented in Figures 8 and 9 only by their weighted average. The individual site declinations near Naselle are of questionable regional tectonic significance because the 90% confidence limits associated with the weighted average of the Naselle sites span the entire range of the Pomona–Pack Sack Lookout declinations. Moreover, the individual site declinations of the Naselle sites are difficult to interpret unambiguously as noted above. Sites MS7 and PO3 were excluded from the regional averaging because their directional reliability is suspect (see above). Site PO4 is plotted in Figures 8 and 9 but was not included in a regional average due to its distance from neighboring sites.

For the 3 to 5 sites included in each regional average, the between-site declination variation is generally consistent with the within-site variation on the basis of χ^2 tests [*Bevington*, 1969]. The only exceptions are averages A and B (Figures 8 and 9) for which the between-site scatter is large enough to suggest that in addition to within-site errors there are also sources of between-site error of the order of 9° (1 σ) for average A and 2° (1 σ) for average B. The cause of the large dispersion in average A has been discussed above. The between-site dispersion of average B is probably due to errors in tilt corrections or paleoanomalies. Local (site specific) paleoanomalies could be expected from buried sources and edge effects associated with paleotopography, such as faults and canyons.

INTERPRETATION

The polarity of both the Pomona Member and the basalt of Pack Sack Lookout is reversed, and the basalts have nearly identical paleomagnetic inclinations (Figure 7), as would be expected for a single flow. The average inclinations (best estimates) differ by only 1° ± 3° (95% confidence), consistent with the correlation of the Pomona with the Pack Sack Lookout basalt. The average declinations differ by 16° ± 5°, a significant difference at the 95% confidence level.

With the possible exception of the five sites near Naselle, we believe that the observed field directions are the consequence of two events. First, approximately 12 m.y. B.P. the Pomona–Pack Sack Lookout basalts erupted as a single flow, cooled, and recorded a nearly uniform magnetization across its total areal extent. Second, at some later time, tectonic processes disrupted the region and rotated the Pack Sack Lookout sites relative to the more easterly Pomona sites. Although a few degrees of the declination discrepancy may result from regional variation of the ancient geomagnetic field, the bulk of the difference can most reasonably be explained as product of tectonic rotation. The small tilts of the flows indicate that most of the rotation took place about a vertical axis. Considering the rate and random nature of secular variation of the earth's magnetic field, the chance of

TABLE 3. Regional Averages of Declination

Label	Sites	Latitude, deg	Longitude, deg	D, deg	σ_μ	S_μ	χ_ν^2
A	WO22, WO50, WO58, WO64, WO78	46.45	236.18	202.5	3.6	17.7*	25.0
B	MS2, MS3, MS4, MS5	46.72	236.50	207.4	2.1	3.6*	3.0
C	MS6, MS10, MS11	46.21	236.82	204.3	2.7*	0.9	0.1
D	PO-5, PO-6, PO-10	45.70	238.62	189.4	9.5	11.2*	1.4
E	PO-1, PO-2, PO-3	46.67	239.44	192.1	6.0*	0.9	0.1
F	PO-7, PO-8, PO-9, PO-11	46.22	239.98	185.8	6.8*	6.4	1.3
G	PO1, PO2, PO-4	46.49	240.96	186.9	3.3	3.5*	1.2
E'	PO-1, PO-2, PO-3, PO-9	46.46	239.53	192.0	4.4*	0.6	0.1
F'	PO-7, PO-8, PO-11	46.36	240.05	183.6	8.0*	7.3	0.8

Labels refer to the regional averages of Figures 8 and 9; sites, sites included in the average; latitude and longitude of location central to all sites; D, weighted average declination = $\Sigma_{i=1}^{N}(D_i/\sigma_i^2)/\Sigma_{i=1}^{N}(1/\sigma_i^2)$, where D_i is declinations of the individual sites, σ_μ is the 90% confidence limits of the mean declination with $\sigma_\mu = t_c[\Sigma_{i=1}^{N} 1/\sigma_i^2]^{-1/2}$, where t_c is the critical value of t distribution with N-1 degrees of freedom, N is the number of sites in the average, σ_i is the angular standard deviation of the individual site declinations; S_μ is the 90% confidence limits of the mean equal to $t_c[1/N - 1\ \Sigma_{i=1}^{N}[1/\sigma_i^2(D - D_i)^2]/\Sigma_{i=1}^{N} 1/\sigma_i^2]^{-1/2}$; χ_ν^2, reduced chi square equal to S_μ^2/σ_μ^2.

*Denotes value used for error bars of Figures 8 and 9.

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two independent flows possessing the same inclination is small and suggests that the cooling times of the Pack Sack Lookout and Pomona units are separated by no more than a few decades. The actual time separation may have been of the order of several days, the time required for the flow to traverse the plateau [*Shaw and Swanson,* 1969]. Data from the five sites near Naselle are also consistent with rotation of the Pack Sack Lookout flow if excessive tilt corrections are the cause of the directional dispersion in the Pack Sack Lookout sills. Alternatively, a combination of rotation about a vertical axis and dispersion due to slow cooling may also explain the Naselle data.

What are the tectonic implications of post 12 m.y. rotation of southwest Washington relative to the Columbia Plateau? As noted in the introduction, there is good reason to believe that the Columbia Plateau of central and eastern Washington has rotated little, if at all, relative to stable North America. If this relation is true, it implies that southwest Washington has undergone a clockwise rotation relative to stable North America. This interpretation is consistent with the well-established larger clockwise rotations of older rocks of the Oregon and Washington Coast Range and western Cascade Range.

Two tectonic models can be advanced to explain the paleomagnetic declinations of the Pomona–Pack Sack Lookout flow, as can a continuum of tectonic models intermediate between these two end-members. The first is a rigid plate model in which southwest Washington rotates as part of an essentially rigid domain separated by a discrete boundary from an unrotated area to the east. The second end-member is a distributed shear model in which rotation accompanies shear distributed throughout the Oregon and Washington region [*Atwater,* 1970; *Beck,* 1976]. Laboratory studies and kinematic models show that shear faulting can produce rotation provided the displacements are large and the distance between faults is small [*Freund,* 1974]. Northwest trending faults along which substantial amounts of shear may have occurred have been recognized in Oregon [*Walker,* 1973; *Lawrence,* 1976] and the Columbia Plateau of Washington [*Bentley,* 1979]. Regional shear should increase to the west and southwest toward the primary North America–Pacific shear boundary, toward the North America–Juan de Fuca oblique convergence boundary, and away from the continental interior, much like the spatial distribution of shear in California and Nevada [*Eaton,* 1980]. The most realistic tectonic model is probably a combination of both the shear and rigid plate models. The paleomagnetic data and the regional geology appear to contain elements of both models and suggest that the rotated domains are of various dimensions.

Character of Rotations in Southwest Washington

Within southwest Washington there appear to be two domains with different styles of rotational tectonics. Within 30 km of the coast, the outcrops of intrusive Pack Sack Lookout basalt give scattered results. In contrast, the area to the east where the Pack Sack Lookout basalt was a subaerial flow yields highly coherent magnetic directions. The different character of the two domains appears to reflect differences in the local geology. All of southwest Washington has undergone deformation which in places strongly folded the mid-Miocene Grande Ronde and Wanapum Basalts prior to emplacement of the Pack Sack Lookout basalt (Figure 2). Deformation continued after 12 m.y. near the coast in the vicinity of Naselle and the Pack Sack Lookout sills. The scattered results from the sills may be due in part to large (10° to 20°) local rotations associated with local, post 12 m.y. structures (Figure 2). Significant local rotation appears likely, especially where faults with horizontal displacement are closely spaced. Uncertainties in the tilt corrections and the possible effects of slow cooling make the results difficult to interpret uniquely.

Toward the east, the subaerial Pack Sack Lookout basalt is much less deformed than in the Naselle region. The magnetization of the Pack Sack Lookout sites near the Columbia River (exclusive of MS7) is virtually identical to that at sites 60 km to the north at Pack Sack Lookout (Figure 2, Table 3). In contrast to the western domain where local rotations may predominate, the eastern domain appears to be associated with rotation of a coherent plate. The declinations of these sites are consistent with a 16° clockwise rotation of a rigid plate with a minimum lateral dimension of 60 km. Rotation could have been a response to distributed west-northwest shear, yet most of the mapped faults cutting the Pack Sack Lookout basalt (Figure 2) appear to have small horizontal displacements (<100 m) where the displacements can be observed or inferred. These small displacements suggest that the west-northwest faults result in minor (<5°?) rotation. Furthermore, the northwest trending faults near the Columbia River are generally absent from the Pack Sack Lookout region, and this absence suggests that the similarity of the paleomagnetic directions between these two regions cannot be explained by nearly equal rotation due to similar amounts of distributed shear. Rotation of a rigid plate with some second-order local rotation (MS7) seems to be the simplest way to account for the paleomagnetic results and to describe the essential elements of deformation in southwest Washington east of Naselle. In summary, coherent regional rotation and complex local rotation both appear to exist, the two being present to different degrees in the eastern and western regions.

Nature of the Transition Between Southwest Washington and the Columbia Plateau

Sheriff and Bentley [1980] propose that much of the Pacific northwest is undergoing regional distributed shear on structures spaced 100 to 1000 m apart. Northwest trending faults recognized in Oregon [*Walker,* 1973; *Lawrence,* 1976] and the Columbia Plateau of Washington [*Bentley,* 1979] may be manifestations of this shear. *Sheriff and Bentley* [1980] have reported a preliminary analysis of the magnetization of the Frenchman Springs Member of the Wanapum Basalt (Columbia River Basalt Group) (Figure 4). These flows are characterized by excursion directions of magnetization, which raises the possibility that the flows formed at a time when the nondipole component of the geomagnetic field was relatively large and the spatial gradients in the field were high. The data of Sheriff and Bentley show a progressive increase of declination from central Washington to the Oregon coast, suggesting a corresponding progressive westward increase in the amount of rotation from 0° in the east to a maximum of 20° in the west. These results are consistent with progressively greater amounts of shear toward the west edge of the North American plate.

On the other hand, the results from the Pomona–Pack Sack Lookout suggest rotation of a rigid plate separated

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from the unrotated region to the east by a relatively sharp boundary, although the existing data lack the resolution to be definitive. The data projection in Figure 8 should be optimal for recognizing a southwesterly increase in declination that might be associated with northwest directed shear. Taken at face value, the regional declination averages do not clearly support the distributed-shear model because no well-defined systematic westerly increase of declination (clockwise rotation) is observed. However, the confidence limits of the averages permit a westerly increase in declination, especially if the trend is more complicated than a simple straight line or is confined to the western 150 km.

A somewhat better fit to the data is exhibited by the rigid plate model (Figure 9) with two tectonic domains, a rotated Pack Sack domain west of the Cascades, and an unrotated plateau domain east of the Cascades. The consistency of the magnetic directions within both domains is impressive. The four regional averages from the plateau domain are consistent with the overall average declination of the plateau sites (190°), and the three Pack Sack Lookout averages are consistent with the overall Pack Sack Lookout declination (205°). At this regional scale the consistency is suggestive of unfragmented coherent plates. However, the data lack the resolution to rule out the existence of small independent blocks rotated 5° to 10° relative to neighboring blocks.

The fit of the shear and rigid plate models to the data can be quantitatively compared using χ^2 statistics. A straight line found by a weighted least squares fit to the data (distributed-shear model) and by a step function (rigid plate model) have a χ^2 value of 193 and 126, respectively, for the N60°E profile (Figure 8) and 187 and 126 for the east-west profile (Figure 9). Weighting was inversely proportional to the standard deviation of the individual data points. Using an F test of the ratio of χ^2 statistics and accounting for the increased degrees of freedom, the rigid plate model (step function) fits the data better with >99% confidence for both cases. Caution is advised in interpreting these statistics because a linear increase in declination is not the only possible shear model. A more complicated relation preserving a westerly increase in declination but including some variability, concentrating the increase of declination in specific zones, would clearly reduce the χ^2 associated with the shear model. Also, the better fit of the rigid plate model is largely dependent on only four data points (PO-5, PO-6, PO-10, and PO4). There is at present no basis to reject these data points. We conclude therefore that the rigid plate model is statistically favored by the data over a shear model with a linear increase in declination. However, more complicated shear models are not excluded by this statistical analysis.

Regional Implications

Two considerations suggest that these results from a small corner of southwest Washington may reflect regional tectonic processes. First, paleomagnetic results from Eocene and Oligocene rocks in the Oregon Coast Range and Oligocene and Miocene rocks in the Oregon and Washington Cascade Range suggest that during the Neogene these terranes were rotating together in a coherent manner [*Beck*, 1962; *Simpson and Cox*, 1977; *Beck and Burr*, 1979; *Magill and Cox*, 1980, 1981; *Bates et al.*, 1981]. We propose that these terranes compose a single Willamette plate bounded on the west by a trench and on the east by the extending Basin and Range province.

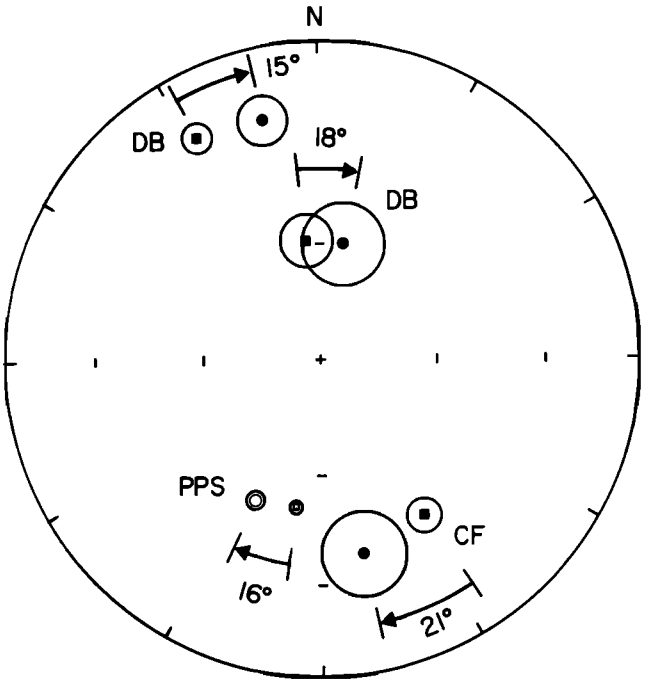


Fig. 10. Stereographic plot showing field directions for sites from the Oregon coastal basalts and probable correlative Columbia Plateau basalts. Note the consistent 15° to 21° clockwise rotation of the coastal units with respect to the plateau. Data are from *Choiniere and Swanson* [1979] and R. W. Simpson and S. D. Sheriff (unpublished data, 1981). Circles are for sites west of the Cascades, and squares are for sites east of the Cascades. DB, Depoe Bay Basalt and flows thought to be correlative from the Grande Ronde Basalt of the eastern Columbia Plateau; CF, Cape Foulweather Basalt and Frenchman Springs Member of the Wanapum Basalt; PPS, Pomona Member of Saddle Mountains Basalt and basalt of Pack Sack Lookout, results from this study. Solid symbols are field directions of normal polarity and open symbols sites of reversed polarity.

Second, intriguing but uncertain paleomagnetic results from Miocene basalts on the Oregon coast agree with the results of this study and suggest that southwest Washington and western Oregon may be rotating as part of the same rigid plate. These basalts, the Cape Foulweather and Depoe Bay basalts, were briefly described earlier. Four distinct directions recorded by the coastal basalts imply 15° to 21° clockwise rotation of the entire coastal region (Figure 10) relative to the eastern Columbia Plateau. These results are uncertain due to large error limits and also because of uncertainties associated with correlation, tilt corrections, and possible nondipole fields. Correlation of the coastal Cape Foulweather and Depoe Bay flows with the plateau Frenchman Springs Member of the Wanapum Basalt and flows of the Grande Ronde Basalt, respectively, is not well established. This is especially true for the Depoe Bay Basalt, which actually consists of several flows with at least four distinct paleomagnetic directions and virtually indistinguishable geochemistry. Thus, one to one correlation with the Grande Ronde Basalt flows is not possible without employing the paleomagnetic data. Uncertainty in these magnetic results is also introduced by uncertainty about the appropriate tilt correction to be used. Because of the intrusive nature of the rocks, adjacent sediments are deformed and in many cases appear to record deformation that occurred at the same time as the intrusion rather than later. For some sites a

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regional correction was inferred from the neighboring geology. Furthermore, three of these four coastal basalts appear to have recorded the geomagnetic field at a time when it was undergoing an excursion in direction, which may be associated with large spatial gradients in the field as discussed earlier. Therefore, some of the differences between the coastal and interior flows may reflect spatial variations in the geomagnetic field rather than tectonic rotations. In view of these uncertainties, we conclude that the coastal basalts of Oregon suggest but do not prove that coastal Oregon and southwest Washington are parts of the same rotating plate.

Whether the southwest corner of Washington has moved with the Willamette plate is uncertain. Structures suggested by gravity and magnetic data appear to separate these areas at about the latitude of the Columbia River, and field mapping of *Wells* [1981] suggests that the course of the Columbia River is in part fault controlled. Furthermore, the fact that the Miocene deformation in the Naselle region was greater compared to the apparent mild deformation of northwest Oregon suggests that the tectonic environments in these two areas may be different. Perhaps southwest Washington and the Willamette plate are only loosely coupled or are rotating independently in response to similar stress fields. Quite possibly the greater deformation in southwest Washington is a boundary deformation at the northern end of the Willamette plate. In any event, we believe that the rotations of southwest Washington cannot be adequately understood without viewing them in a context ultimately encompassing much of the western United States.

Magill and Cox [1980, 1981] have proposed that extension of the Basin and Range province [*Stewart,* 1978; *Zoback et al.,* 1981] was coincident with and tectonically related to the clockwise rotation of the Willamette plate and argue that the low heat flow and absence of faulting indicate a nearly rigid rotation of the plate. Moreover, *Zoback et al.* [1981] propose that 20 to 30% extension of the Basin and Range province has occurred during the past 11 m.y., notably after 12 m.y. B.P., when the Pomona–Pack Sack Lookout basalt erupted and cooled. An extension of the Basin and Range province of 25% at latitude 41°N and decreasing to the north would produce 15° clockwise rotation of the Willamette plate using the tectonic model and Euler pole of *Magill and Cox* [1980]. Hence, a post 11 m.y. clockwise rotation of 16° is consistent with a regional tectonic model in which southwest Washington rotated as part of a larger Willamette plate.

The Neogene Cascades: A Tectonic Boundary?

The paleomagnetic data from the Pomona Member and the basalt of Pack Sack Lookout suggest that if a tectonic boundary exists between the Pack Sack and plateau domains (Figures 3 and 9), it must lie within a 120-km gap occupied by the Cascades. Because data are absent within the gap, we cannot determine whether this boundary is a broad transition zone or a sharp boundary. We propose that during the past 12 m.y., and possibly throughout the Neogene, the Cascade Range from southern Washington to northern California (which we will refer to as the southern Cascades) has been a major tectonic boundary separating two domains: the Basin and Range province, an extensional domain characterized by translation toward the northwest, and the clockwise-rotating Willamette plate, a more rigid domain consisting of the western Cascades and Coast Ranges of Oregon and southern Washington.

Clues to the nature and extent of the Cascade tectonic boundary may be found in changes in the style of volcanism, geologic structure, and geophysical characteristics along the length of the Cascade arc. From Mount Rainier, Washington, northward into British Columbia, the arc consists of large isolated andesitic and dacitic stratovolcanoes resting on relatively thick pre-Tertiary crust. In contrast, the Cascade Range south of Mount Rainier in Washington, Oregon, and northern California is marked by thinner crust [*Smith,* 1978] and much more extensive volcanism (Figure 11), with basalt volumetrically more important than andesite [*White and McBirney,* 1978]. The abundant basaltic volcanism in the southern Cascades can be spatially and temporally correlated with the formation of a discontinuous axial graden, at least during the last 5 m.y. [*Taylor,* 1980, 1981; *McBirney,* 1978]. Although central grabens are common in volcanic arcs [*Williams and McBirney,* 1979], the special significance of the extensional volcanism and structure in the southern Cascades lies in their juxtaposition with the Basin and Range province. The discontinuous axial graben of the southern Cascades appears as en echelon segments that widen and merge southward into the extensional Basin and Range province [*Lawrence,* 1976; *Hammond,* 1979]. This could be interpreted as an invasion of basin and range extensional tectonics into the arc, with extension pinching out northward along the axis into central Washington, where the crustal thickness and velocity structure are more typical of nonextensional regimes [*York and Helmberger,* 1973; *Blackwell,* 1978; *Lachenbruch and Sass,* 1978; *Smith,* 1978]. We envision the southern Cascades as the leading edge of

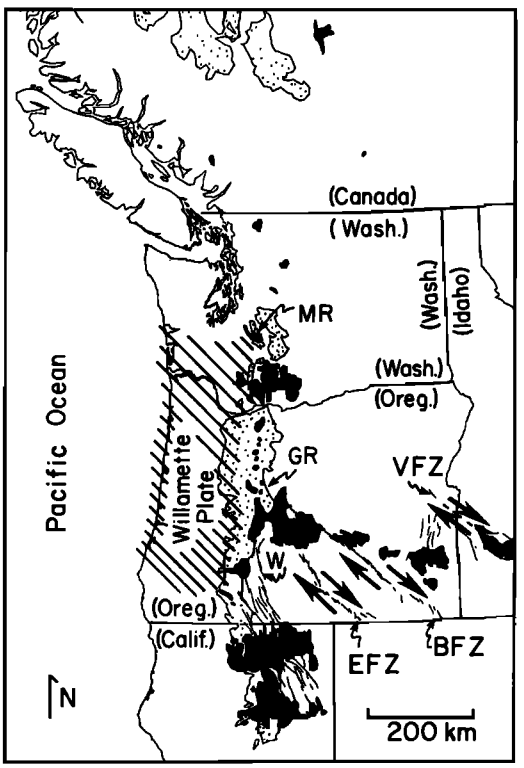


Fig. 11. Generalized geologic map of the Pacific northwest showing Quaternary volcanism in black and post 17 m.y. Cascade volcanism in stippled pattern. EFZ, Eugene-Denio fault zone; BFZ, Brothers fault zone; VFZ, Vale fault zone; MR, Mount Rainier; GR, Green Ridge; W, Walker Rim. Geology from *Douglas* [1968], *King and Beikman* [1974], and *Stewart* [1978].

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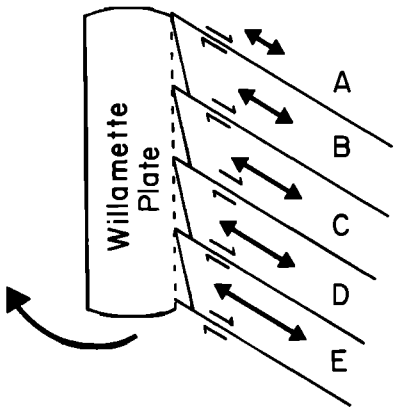


Fig. 12. Cartoon of the Basin and Range province characterized by NW translation meeting the eastern boundary of the more rigid Willamette plate. Northward termination of extension (shown by arrows with region A having the least extension and region E the greatest extension) would produce transform structures and clockwise rotation of the plate. Displacement discontinuities at the eastern boundary (coincident with the Neogene Cascades) produce extensional zones prone to volcanic activity.

the extensional Basin and Range province acting as a buffer between the differentially expanding Basin and Range province and the rotating, nearly rigid terrane to the west. The geometry of this relation is shown schematically in Figure 12 for an ideally rigid Willamette plate rotating in response to differential extension and translation in the Basin and Range province. The actual geometry of interaction is probably much more complex than pictured here, but it is interesting to note that the en echelon triangular grabens predicted along the boundary may exist in Oregon where major northwest-trending normal faults such as Green Ridge [*Lawrence*, 1976] and Walker Rim (Figure 11) intersect the axis of the Cascade arc. Structural relations along the west side of the boundary are not well known. A more complete understanding of the importance of the Cascades as a tectonic boundary is dependent on much needed geologic mapping in the western Cascades.

Driving Forces

The merging of extensional basin and range faulting with the Neogene Oregon Cascades (Figure 11) may have several tectonic implications. The absence of conspicuous folding on the east edge of the Willamette plate indicates that the plate is not simply pushed from the east along a localized front. Yet Neogene rotation of the Willamette plate could occur in much the same manner as oceanic crust is pushed away from a mid-ocean ridge without being deformed [*Jacoby*, 1970; *Bott and Dean*, 1972; *Artyushkov*, 1973, 1974]. The high topography and thin lithosphere of the Basin and Range province have been modeled as a diffuse oceanic ridge [see *Suppe et al.*, 1975; *Eaton*, 1979]. The Willamette plate could be receiving a ridge-push force from the Basin and Range province; the plate essentially sliding westward off the topographic and asthenospheric high to the east, similar to a massive gravity slide with the base of the sliding plate being below the elastic layer and probably coincident with the base of the lithosphere. The clockwise sense of the rotation would be a consequence of the northward reduction of extension in Oregon and Washington [*Eaton*, 1979] and the associated northward reduction of the ridge-push force. Interestingly, a push from the Basin and Range province should exist in all directions away from the high topography at the center of the province. The Miocene and younger east-west fold axes of the Columbia Plateau [see *Davis*, 1977] may result from a northward push exerted by the Basin and Range province against a more rigid continental buttress in northern Washington. No buttress exists to the west, where the Willamette plate simply overrides the Juan de Fuca trench.

Alternatively, *Beck* [1976, 1980] has proposed that many of the clockwise rotations documented along the west coast of North America are a consequence of ball-bearing tectonics, whereby small plates are caught in the shear couple between the North America and Pacific plates. The Juan de Fuca plate offshore of Oregon and Washington may have rotated by this mechanism about 10° in the last 10 m.y. judging by the fanning of the marine magnetic anomalies [*Atwater and Menard*, 1970; *Hays and Pitman*, 1970]. This rate of rotation is comparable to the rates observed onshore and may reflect a degree of coupling and a similar causal mechanism for both onshore and offshore rotations. The ridge-push and ball-bearing mechanisms may well be operating in conjunction in the Pacific northwest, but our present understanding of cause and effect does not allow us to resolve the effects of each.

CONCLUSIONS

Nearly identical paleomagnetic inclinations from the Pomona Member and the basalt of Pack Sack Lookout are consistent with the stratigraphic, geochemical, mineralogic, and geologic data identifying the Pomona and Pack Sack Lookout as parts of the same flow. Furthermore, the magnetic declinations establish that southwestern Washington has rotated ~16° clockwise with respect to the Columbia Plateau during the past 12 m.y. The amount and sense of this rotation are consistent with older rotations reported from the Coast Ranges and western Cascades of Oregon and Washington.

The style of regional rotation is not determined perfectly by the present data because of uncertainties in tilt correction and local tectonic effects. However, the data suggest that the Cascades act as a tectonic boundary between the rotated coastal region and the unrotated Columbia Plateau. The lithosphere under the Cascade Range may be hotter and weaker than the lithosphere in neighboring regions and would be a likely place for a major tectonic discontinuity. This interpretation would favor rigid plate rotation rather than the distributed-shear rotations described by *Sheriff and Bentley* [1980]. Sampling of older flows between the plateau and southwest Washington should help resolve this question.

Deformation in the rotated plate is probably a mixture of rigid plate rotation and distributed shear. The relative effects of these two processes are determined at any site and time by conditions such as the thermal state of the lithosphere and orientation of preexisting fractures. Evidence seems to favor a dominantly rigid plate rotation of the Willamette plate and southwest Washington. We envision the rotation of these regions to be occurring as a necessary kinematic counterpart to extension of the Basin and Range province. If this model is correct and the entire Willamette plate did rotate during the past 12 m.y., 25% extension of the Basin and Range province at latitude 41°N is required. This model further suggests that during the Neogene the Cascades from central

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Washington to northern California formed the eastern tectonic boundary of the Willamette plate, with the southern Cascades accommodating differential displacements between the translating and extending Basin and Range province and the more rigid rotating Willamette plate. In a broader context, the rotations in the Pacific Northwest may be caused by a ridge-push force originating in the Basin and Range province acting alone or in conjunction with shear forces along the Pacific–North America plate boundary.

APPENDIX: SITE LOCATIONS—BASALT OF PACK SACK LOOKOUT

MS2. Raymond 15′ Quadrangle, Washington (1955). Center of NW1/4 of section 10, R7W, T14N. About 3.8 km SSE of Pack Sack Lookout. Quarry at top of road E1000 from A Line–G Line junction. Tilts: local sediments 310, 16NE; regional topography 339,6NE; structure from map of *Wagner* [1967] 332, 7NE.

MS3. Raymond 15′ Quadrangle, Washington (1955). SW1/4 of SW1/4 of section 13, R7W, T14N. About 7.6 km SE of Pack Sack Lookout in quarry west of road E1000, just west of intersection with D Line. Tilts: local sediments below flow 325,4NE; regional topography 292, 5NNE; structure from map of *Wagner* [1967] 313,12NE.

MS4. Pe Ell 15′ Quadrangle, Washington (1953). SE1/4 of SW1/4 of section 1, R7W, T14N. About 5.5 km ESE of Pack Sack Lookout, quarry on E side of D Line just north of intersection with F Line. Outcrop 15 to 20 m high with good columns; vesicular toward top and capped by sediments. A small low-angle fault cuts outcrop, dips to south (samples from below fault). Tilts: local sediments above basalt in quarry to south 350,6E; regional topography 341,4NE; structure from map of *Pease and Hoover* [1957] 29,7SE.

MS5. Pe Ell 15′ Quadrangle, Washington (1953). NW1/4 of NW1/4 of section 5, R6W, T14N. About 6.8 km ESE of Pack Sack Lookout. About 30 m long outcrop of columnar basalt along N side of road, about 100 m E of where road crosses S flowing stream, 2 to 2.5 m thickness exposed. Tilts: on columns 60,5SE; Structure from map of *Pease and Hoover* [1957] 35,4SE.

MS6. Clatskanie 15′ Quadrangle, Oregon-Washington (1952). Top center of NW1/4 of section 7, R4W, T8N. South 0.4 km from Cathlamet Road, about 12.2 km E of Cathlamet, quarry exposure of large columns. Tilts: columns are approximately vertical; regional topography about 330,2NE.

MS7. Cathlamet 15′ Quadrangle, Oregon-Washington (1953). SW1/4 of SE1/4 of section 3, R5W, T8N. About 7.7 km E of Cathlamet, in quarry. Massive basalt with platey partings near top of flow. Samples from approximately 5m below top of flow. Tilts: from partings at top of flow, ~270,6N; from regional topography 330,2NE.

MS10. Clatskanie 15′ Quadrangle, Oregon-Washington (1952). Center of SE1/4 of section 36, R5W, T9N. About 10.9 km east of Cathlamet, in quarry on S side of logging road, about 7m high exposure of basalt with wavy columns. Top and base of flow are not visible. Tilts: columns are roughly vertical; regional topography 330,2NE.

MS11. Clatskanie 15′ Quadrangle, Oregon-Washington (1952). NE1/4 of NE1/4 of section 6, R3W, T8N. Quarry on Harmony Creek Road, about 3.4 km NE of Stella, samples from v-shaped end of quarry to southwest. 10 m of wavy columns exposed. Tilts: columns are roughly vertical.

WO22. Oman Ranch 7.5′ Quadrangle, Washington (1949). SE1/4 of section 17, R9W, T11N. Quarry on crest of Radar Ridge. Tilts: local sediments 330,15NE, 340,17NE, 310,21 NE.

WO50. Oman Ranch 7.5′ Quadrangle, Washington (1949). SW1/4 of NW1/4 of section 3, R9W, T11N. In quarry on NE limb of Radar Ridge syncline, platey jointed, overlain by baked siltstone. Tilts: basalt foliation 115,32S; siltstone above sill 100,28S; sediments below sill 0.8 km away 120,15S.

WO58. Oman Ranch 7.5′ Quadrangle, Washington (1949). SE1/4 of SE1/4 of section 10, R9W, T11N. Small outcrop on NE limb of Radar Ridge syncline along logging road. Tilts: basalt jointing 130,23SW; sediments above sill 0.4 km from site 130,29SW to 140,29SW.

WO64. Oman Ranch 7.5′ Quadrangle, Washington (1949). SW1/4 of SW1/4 of section 31, R9W, T11N. Quarry on E limb of Nemah R. syncline on Middle Nemah C Line. Tilts: basalt foliation 165,49W in lower quarry and 170,53W in upper quarry; nearest siltstone 165,36W to 165,46W (separated from site by fault).

WO78. Long Island 7.5′ Quadrangle, Washington (1949). Center of Section 10, R10W, T11N. West limb of Nemah River Syncline, quarry on north side of logging road. Tilts: basalt jointing 320,40N with dip variable up to 50°+; Siltstone below sill 320,20NE to 320,30NE.

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